07 Changes in Concentration Due to Conversion

Why are we learning this?

In previous lectures, we rewrote mole balances in terms of conversion. However, reaction rates are written as functions of concentration. To solve reactor design problems, we need a way to express concentration in terms of conversion.

By the end of today's lesson, you should understand:

- How conversion affects moles and molar flow rates.
- How stoichiometry relates all species to the limiting reactant.
- Why concentrations change as conversion increases.
- Why gas-phase systems can experience volume changes.
- Why liquid-phase systems are often treated as constant volume.
- How to express concentration as a function of conversion.

Course Roadmap

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(Conversion)
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Equilibrium
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Equilibrium Conversion
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►One Reaction: Solving in terms of Reaction Progress◄
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Challenging the Well-Mixed Assumption
(Residence Time Distributions)
↓
Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Big Idea

To solve reactor design equations, we ultimately need:

Rate Law
↓
Concentration
↓
Conversion

Today's lecture builds that connection.

In this lecture, we'll write concentration in terms of conversion. Concentration is written as:

- In a flow reactor,

- In a batch reactor,

Conversion can influence *F*, *N*, *V*, and υ.

Why Do We Care About Concentration?

Most rate laws are written in terms of concentration:

-rA = k*f(CA, CB, ...)

Reactor design equations are often written in terms of conversion.

To combine the two, we must express concentration as a function of conversion. This is one of the most important connections in reactor design.

First, let's recall the influence that *X* has on *F* and *N*. This is review from Material and Energy Balances:

Consider a reactor carrying out the reaction aA + bB --> cC + dD:

Don't Forget the Signs

For stoichiometric coefficients:

- Reactants → Negative
- Products → Positive

This sign convention ensures that species are consumed and generated correctly when conversion increases. Many algebra mistakes later in the semester come from forgetting this convention.

Let's say that species A is the limiting reactant.

- Then, we can solve the mole balances for these reactors in terms of X_A.
- We can also write values of N and F for species B, C, and D in terms of X_A:

For more on this, see Tables 4-1 and 4-3 in Fogler and also watch the video in today's module on constructing stoichiometric tables. See also Fogler Example Problem 4-1.

Re-arranging these formulas:

Important quantity/definition: $\frac{N_{i0}}{N_{A0}}$ or $\frac{F_{i0}}{F_{A0}} = \Theta_i$

$$\Theta_i = \frac{\text{moles of species } i \text{ initially or moles of species } i \text{ fed}}{\text{moles of species A initially or moles of species A fed}}.$$

Hence, $N_i = N_{A0}\Theta_i \left(\frac{\nu_i}{\nu_A}\right) N_{A0}X_A$ and $F_i = F_{A0}\Theta_i \left(\frac{\nu_i}{\nu_A}\right) F_{A0}X_A$, where ν are stoichiometric coefficients.

Meaning of Θ_i

Θ_i compares the amount of species *i* to the amount of limiting reactant present initially (or fed).

$$\Theta_i = \frac{\text{Species i initially/fed}}{\text{Limiting reactant initially/fed}}$$

Think of Θ_i as simply a convenient bookkeeping tool for describing feed composition.

Stoichiometry Is Doing the Heavy Lifting

Conversion tells us how much of the limiting reactant has been consumed. Stoichiometry then tells us how all other species must change. Once we know:

- The relative amounts fed/present initially,
- the reaction stoichiometry, and
- the conversion,

we can determine the amount of every species present in the reactor.

<u>Important note:</u> For proper accounting, stoichiometric coefficients are negative for reactants and positive for products.

<u>Another important note:</u>

- We can always write conversion for any of the reactant species, even if they are not the limiting reactant. For example, $X_A = \frac{F_{A0}-F_A}{F_{A0}}$, and $X_B = \frac{F_{B0}-F_B}{F_{B0}}$. These quantities tell us the fraction of species A and B, respectively, that has reacted.

- However, if we want to solve a mole balance in terms of *X*, we must use the conversion of the limiting reactant.

- We write most of our definitions and balances in this class in terms of species A … However, species A isn't always the limiting reactant. <u>Always use the limiting reactant as your basis of calculation when solving mole balances in terms of *X*</u>.

Why Use the Limiting Reactant?

Conversion can be defined for any reactant. However, when solving mole balances in terms of conversion, the limiting reactant must be used as the basis. Why?

Because the limiting reactant uniquely tracks the progression of the reaction. Using a non-limiting reactant can let the reaction proceed past the point of depletion of the limiting reactant.

Now let's see how X influences V and υ.

Consider the gas phase reaction A --> B + C, and for the sake of illustration, imagine that this reaction is being carried out in a balloon:

The balloon has non-rigid walls.
Assume that the reaction is carried out isothermally and that
the reaction goes to completion.

The balloon is essentially a constant pressure batch reactor.

Let's consider the case where only species A is present initially and the reaction goes to completion, so ultimately, the balloon only contains species B and C.

At time = 0:

When the reaction has gone to completion:

Why Might Volume Change During Reaction?

As a reaction proceeds, the total number of moles may change.

Example:

$A \rightarrow B + C$

One mole becomes two moles. If pressure and temperature remain constant, additional moles require additional volume. This is why many gas-phase systems experience expansion or contraction during reaction.

Let's look at the analogous circumstance in a flow reactor:

Gas phase reaction, isothermal, Isobaric, reaction goes to completion.

Gas-Phase and Liquid-Phase Reactions Behave Differently

Gas-Phase Reactions: Changes in total moles often produce significant changes in:

- Volume
- Volumetric flow rate

Liquid-Phase Reactions: Changes in total moles usually have little effect on:

- Volume
- Volumetric flow rate

For most liquid-phase reactor problems in this course, volume can be assumed constant.

Why Does Expansion/Contraction Matter?

Reaction rate depends on concentration.

Concentration depends on volume.

Concentration changes during a reaction for two reasons:

1. Species are consumed or produced.
2. The reactor volume or volumetric flowrate changes.

Ignoring volume/volumetric flowrate changes can therefore lead to incorrect reactor design calculations.

Summary and Key Takeaways

Today we developed the relationships needed to express concentration as a function of conversion.

Key Ideas

☑ Conversion determines how the amounts of all species in a single reaction change through stoichiometry.

☑ The limiting reactant should be used when solving mole balances in terms of conversion.

☑ Feed composition can be represented using Θ values.

☑ Stoichiometric coefficients are negative for reactants and positive for products.

☑ Gas-phase reactions may experience significant volume or volumetric flow-rate changes as conversion increases.

☑ Liquid-phase reactions can often be treated as constant volume/volumetric flowrate systems.

Most Important Idea

Rate Law
↓
Concentration
↓
Conversion

If we can write concentration as a function of conversion, then we can express the rate law in terms of conversion and solve simple reactor design equations.